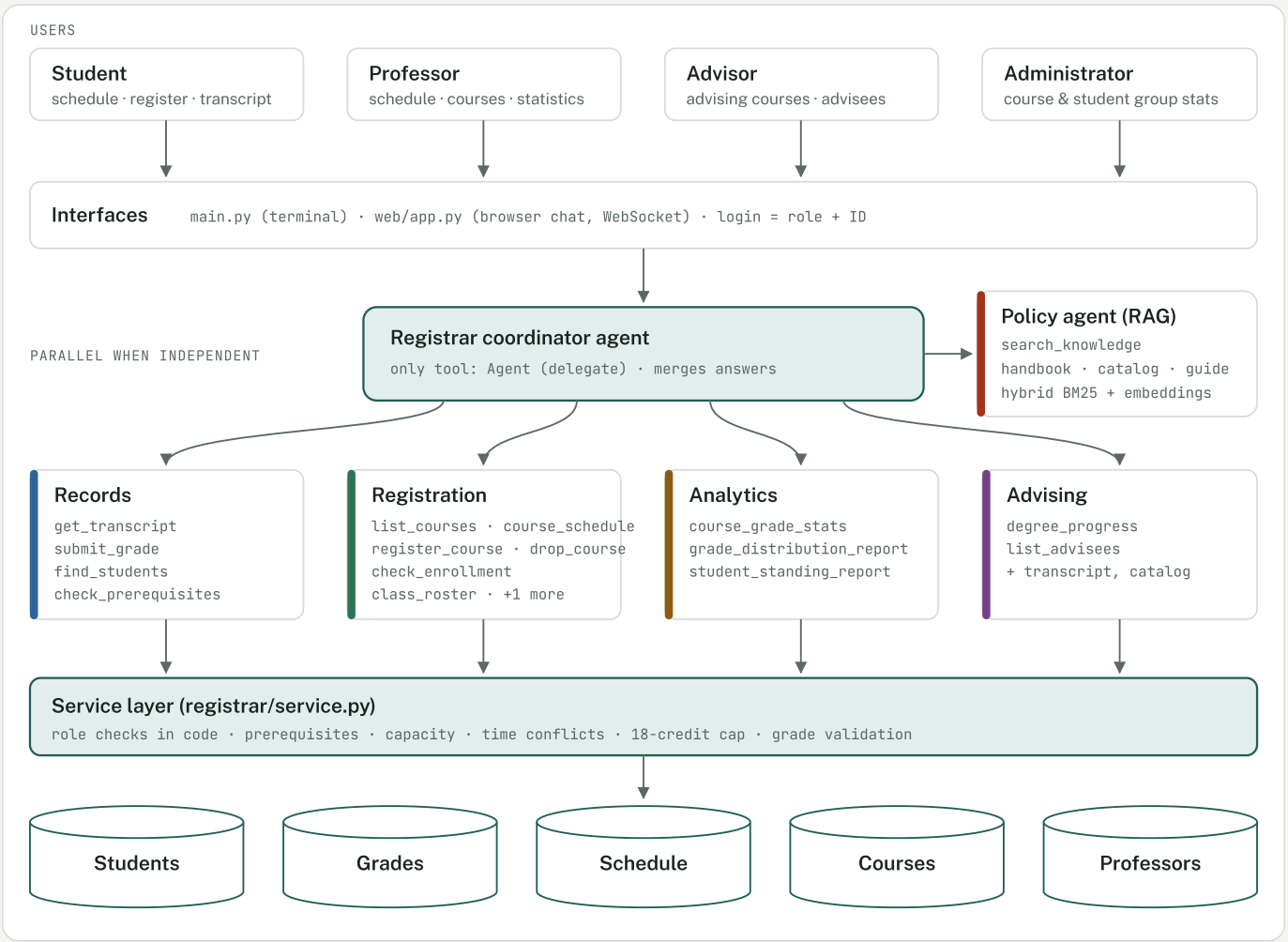
PROJECT ARCHITECTURE · CLAUDE AGENT SDK · PYTHON

Agentic AI Registrar Office

A multi-agent system that acts like a university registrar office. It keeps five databases and answers students, professors, advisors and administrators. Each user only gets the information they're allowed to see.

System architecture

The Registrar coordinator can't read any data. It routes each request to one of four specialist agents, sending independent parts to several at once, and merges their answers. Specialists reach the data only through tools, and every tool goes through one service layer that checks the user's role.



Tools are filtered twice. Each specialist only gets the tools for its own area. Then any tool the logged-in role can't use is removed before the session starts. A student session, for example, has no Analytics agent at all.

Agents working together: an enrollment, traced

Enrollment is a chain. Each specialist does one step, and the Registrar hands its findings to the next agent, so each one builds on what the others found. This is a real run from testing, as student S1001:

“My ID is S1001. Please enroll me in Artificial Intelligence (CS310) for Spring 2027.”

1. **Records** calls `check_prerequisites(CS310)`
   CS310 isn't completed yet, and its prerequisite CS201 is completed with a C, which meets the C-minimum.

2. **Registration** receives Records' findings and calls `check_enrollment(CS310)`
   35 of 35 seats open, no time clash with the student's other Spring 2027 courses, and 3 credits against the 18-credit limit.

3. **Advising** receives both findings and calls `degree_progress`
   CS310 is an elective that counts toward the 42 credits; with 34 credits earned and no core courses left, 5 credits would remain after it.

4. **Registration** calls `register_course(CS310)`
   The service layer runs every check again in code before saving, so the agents' word is never trusted alone. Enrolled: MWF 13:00–13:50 with Dr. Christian Wayllace.

5. **Registrar** writes one reply with a “How we checked” list
   One line per agent with what it found. When a step fails, the chain stops there: asking for a course whose prerequisite is still in progress stops at Records, and nothing is registered.

Knowledge retrieval (RAG)

Rules, deadlines and course descriptions live in the registrar's documents (`knowledge/`): the 2026-2027 handbook, a superseded 2024-2025 handbook, the course catalog and an advising guide. They are split into 41 section-sized chunks, embedded with a small local model and searched with hybrid retrieval (BM25 keyword search plus embedding search, combined by reciprocal rank fusion). The **Policy agent** answers from the retrieved passages with citations, prefers current rules over superseded ones, searches again when results are poor, and says when the documents don't cover a question. The Advising agent uses the same search, and a refused enrollment step is explained with the handbook rule behind it. Full details and evaluation: `docs/RAG_Report.pdf` .

Who can see what

These rules are checked in code on every tool call. The prompts describe them, but the prompts don't enforce them.

CAPABILITY	STUDENT	PROFESSOR	ADVISOR	ADMINISTRATOR
Course catalog and semester schedule	all	all	all	all
Transcript, GPA, degree audit	own only	no	advisees	any student
Register / drop courses	own, Spring 2027	no	no	any student
Teaching schedule and class roster	no	own sections	own sections	all
Submit grades	no	own sections, Fall 2026	own sections	no
Grade stats for one course	no	terms they taught	terms they taught	all
Course-group and student-group reports	no	no	no	all

Data model

These are the five tables from the project brief. A few columns were added so the features work: prerequisites for registration, an advisor for each student, and seat capacity and meeting times for schedules. A grade of `NULL` means the student is enrolled and the course has no grade yet. Computer Science courses are taught by real NMSU Computer Science faculty, matched to courses by research area; their schedules, students and grades are fictional sample data.

Students

id S1001…S1060
name
major
advisor_id → Professors
entry_semester

Grades

student_id
course
semester
grade A … F, or NULL

Schedule

course
semester
professor_id
days · time · room
capacity (35)

Courses

code CS101…PHYS101 (22 courses)
title
credits 3 or 4
prerequisites

Professors

id P01…P13
name
department · title

Registrar policies in the sample data

42 credits

to graduate, with all 8 core courses and a GPA of at least 2.0

GPA < 2.0

marks a student as at risk of dismissal. A GPA from 2.0 to 2.3 is probation watch

C- or better

counts toward a prerequisite

18 credits

is the most a student can take in one semester

Fall 2026

is in progress, and registration is open for Spring 2027

Code: `registrar/db.py` · `registrar/service.py` · `registrar/agents.py` · `main.py` · `web/app.py`